

Subject Description Form

Subject Code	ITC2202T
Subject Title	Foundations of Textiles I
Credit Value	3
Level	2
Pre-requisite/ Co-requisite/ Exclusion	Exclusion: SFT202FY Introduction to Fashion Materials
Objectives	The subject provides the fundamental, professional and technological knowledge of fashion materials including textile fibres, yarns and fabrics.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) understand different types of fibre materials needed for fashion; (b) understand the formation principle of textile yarns; (c) explain the principal methods of yarn production and relate them to the general types of yarn and their properties; (d) compare the structure and performance of different types of textile yarn; (e) describe and explain the formation principles of basic woven and knitted fabrics, and identify the machines for weaving and knitting processes; (f) understand and describe the basic fabric structures and properties with respect to their end-uses.
Subject Synopsis/ Indicative Syllabus	(I) Fibers, Yarns and Their Formation Techniques Types of textile fibers; classification, structure, properties and uses of textile yarns; principles underlying the processes used in the conversion of fibres to yarn; yarn spinning systems; yarn count and yarn twist. (II) Weaving and Woven Fabrics Basic formation principles of woven fabrics; introduction of weaving looms; representation of a woven fabric; classification of woven fabrics; structure, properties and end-uses of woven fabrics. (III) Knitting and Knitted Fabrics Loop basic formation principle; introduction of knitting machines; representation of a knitted fabric; classification of knitted fabrics; structures, properties and end-uses of knitted fabrics.

Teaching/Learning Methodology	Teaching will primarily be conducted in lectures. Appropriate demonstration and analysis of fibre, yarn and fabric samples as well as related instruments and manufacturing machines will be conducted in the laboratories. Online learning materials will also be incorporated https://cottonuniversity.org/courses/online/ Sourcing & Manufacturing: • Fibers Science: Natural vs. Synthetic Fibers; Cotton Fiber Harvesting & Ginning • Yarn Manufacturing: Basic of Yarn Manufacturing; Yarn Spinning • Weaving: Weaving Basics; Woven Fabric Designs; Types of Weaves; Basic Function of the Weaving Loom • Knitting: Knit Basics; Designing Knits; Single Knits; Double Knits																																																						
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>Continuous Assessment</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td><i>1. Assignment</i></td><td><i>30%</i></td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td><i>2. Project/Report</i></td><td><i>20%</i></td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Examination</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: This subject will provide fundamental theory and basic laboratory experience to students in understanding textile materials needed for fashion. Coursework will be used to assess students’ learning outcomes at different stages and aspects of the subject. A final examination will be used to assess the overall understanding of the subject contents. The materials submitted for all assessment must be the student’s own work. The submitted work may not be accepted for the purpose of assessment if its authenticity is questionable. Submitting	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d	e	f	Continuous Assessment	50%	✓	✓	✓	✓	✓	✓	<i>1. Assignment</i>	<i>30%</i>	✓	✓	✓	✓	✓	✓	<i>2. Project/Report</i>	<i>20%</i>	✓	✓	✓	✓	✓	✓	Examination	50%	✓	✓	✓	✓	✓	✓	Total	100%						
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	GenAI generated materials as students' own work or part of their work is an act of academic dishonesty. Students who are found committing academic dishonesty will face disciplinary actions.	
Student Study Effort Expected	Class contact:	
	• Lecture	26 Hrs.
	• Laboratory	12 Hrs.
	Other student study effort:	
	• Self-study	67 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	<u>Books</u> Adanur S. (2001), <i>Handbook of Weaving</i>. Technomic Pub. Co. Inc. Collier, B.J., Bide, M.J., Tortora, P.G. (2009), <i>Understanding Textiles</i>. Prentice Hall, New Jersey. Elsasser, V.H. (2010), <i>Textiles: concepts and principles</i>. Fairchild Publications, New York. Kadolph, S.J., Marcketti, S.B. (2017), <i>Textiles</i>. Pearson Higher Education, USA. Laurence, C.A. (2003), <i>Fundamentals of Spun Yarn Technology</i>. CRC Press, Florida. Shaikh I. A. (2005), <i>Pocket Knitting Expert</i>. Textile Training Books, Textile Info Society. Shaikh, I. A. (2005), <i>Pocket Weaving Expert</i>. Textile Training Books, Textile Info Society. Hann, M. (2021), <i>Textile Design, products and process</i>, CRC Press, Florida	